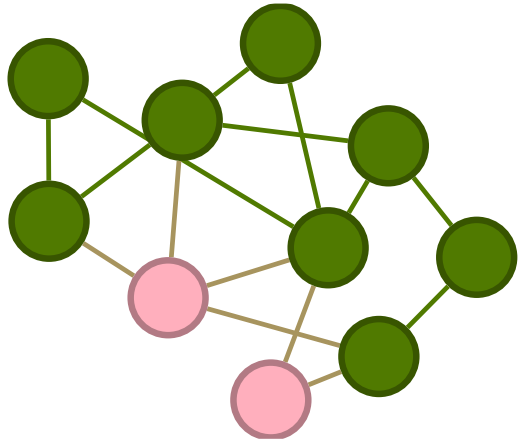
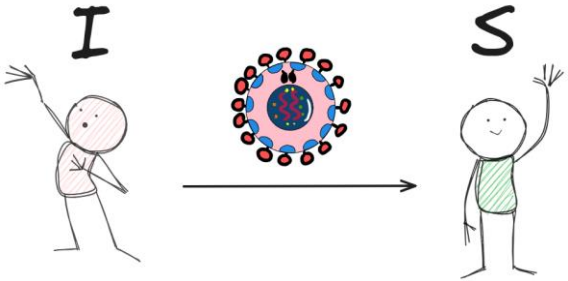
BASIC REPRODUCTIVE NUMBER IN FINITE POPULATIONS



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Basic Reproductive number, R_0

Expected number of cases directly generated by one case in an infinite population where all individuals are susceptible to infection.

WHY IS IT USED?

It is supposed to include all the epidemic parameters.

$R_0 > 1$ The epidemic percolates in the population

$R_0 < 1$ The epidemic **does not** percolate in the population

Basic Reproductive number, R_0

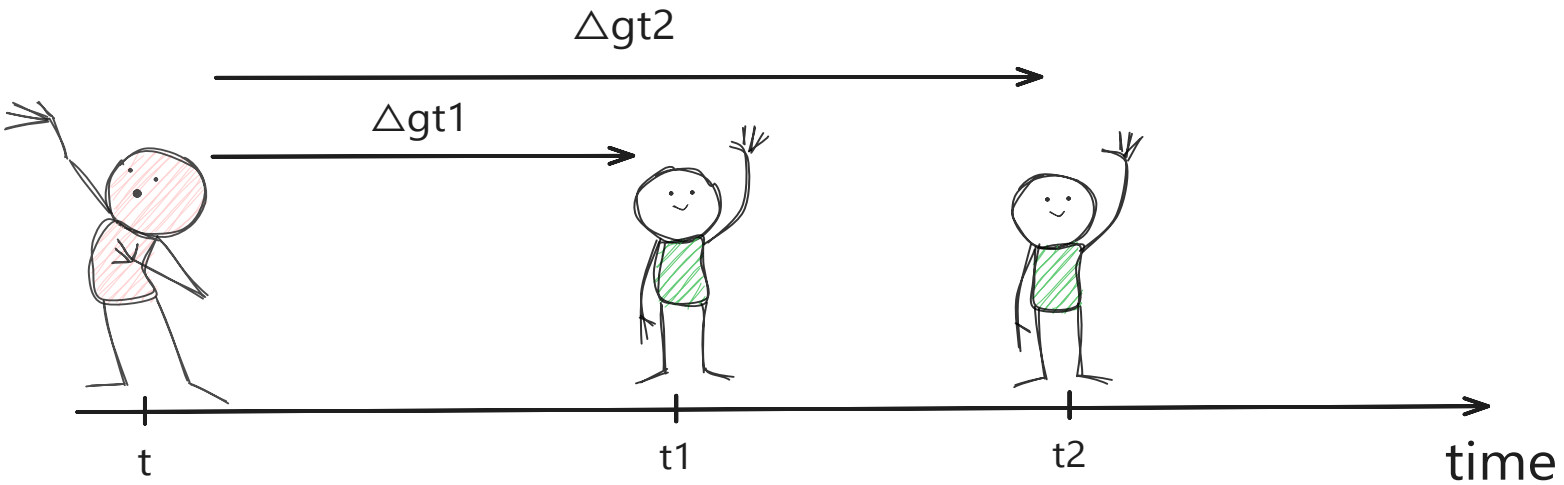
Expected number of cases directly generated by one case in an infinite population where all individuals are susceptible to infection.

HOW CAN WE ESTIMATE IT?

In a real epidemic, it is estimated using the **generation time** at the beginning of the diffusion.

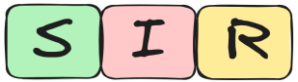
Generation time

It is the time difference between an infected individual becoming infected and their secondary infections.

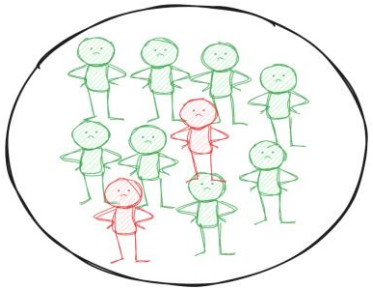


Simulation Process

Assumptions:



- SIR model.
- Homogeneous mixing.
- Theoretical values from Hiroshi Nishiura 2010 [1].



Algorithm:

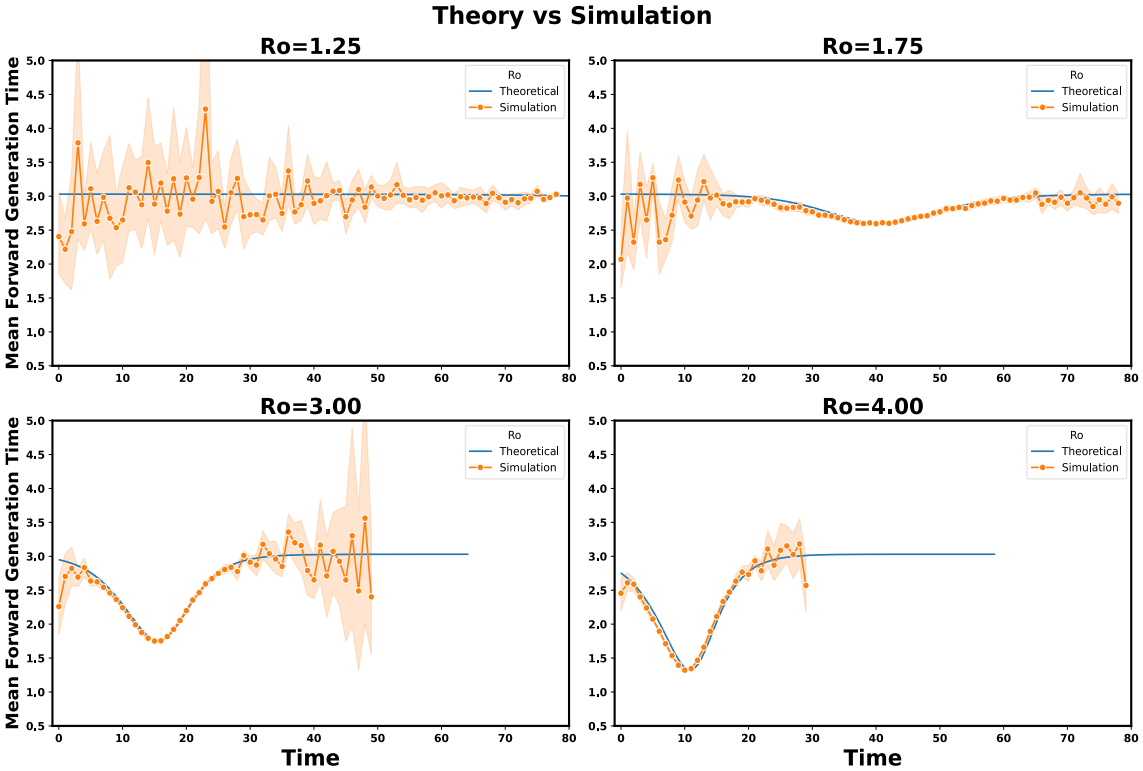
- Gillespie algorithm.

Discrete time → Continuous time

[1] Nishiura, H. (2010). Time variations in the generation time of an infectious disease: Implications for sampling to appropriately quantify transmission potential. Mathematical Biosciences & Engineering, 7(4), 851-869.

Generation time results

- We need to use the initial generation time.
- For bigger values of R_0, we have a drop in these values due to a susceptible depletion.
- It is not possible to use the initial generation time for $R_0 > 3$.

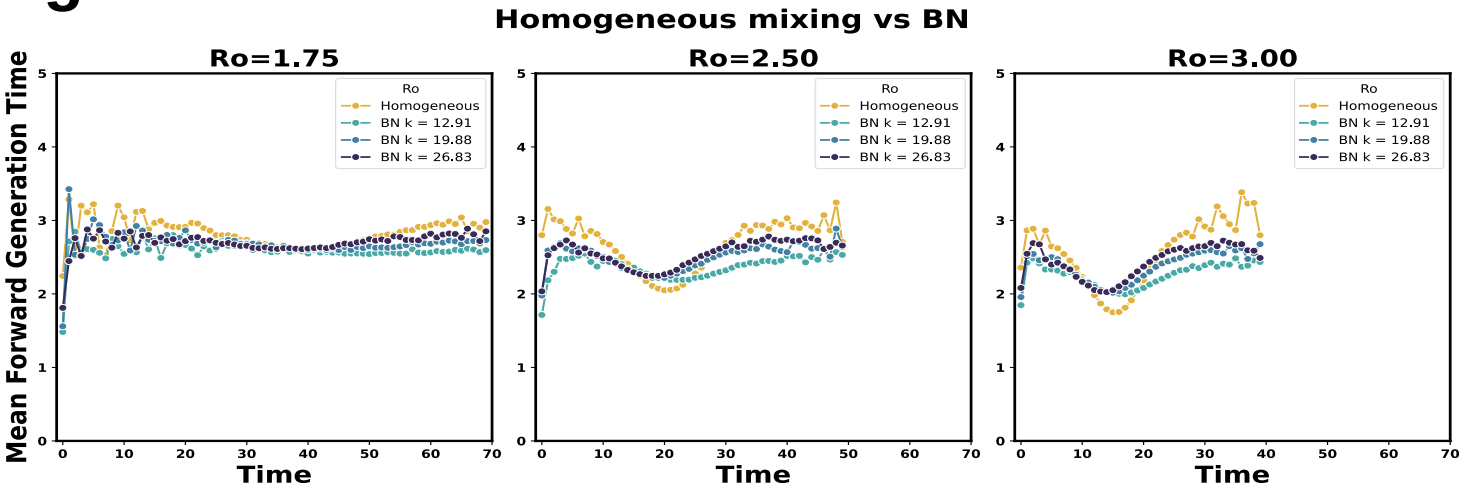


R_0 estimation

- Package R0 of R (2023).
- Attack rate is a theoretical method for SIR model.
- Very poor estimations for $R_0 > 2$.

R_0	Method		
	Attack rate	Exponential growth	Maximum Likelihood
1.25	1.24	1.20	1.22
1.75	1.73	2.38	2.10
2.50	2.48	2.03	1.57
3.00	2.98	1.21	1.05
4.00	3.97	0.91	1.00

Adding networks



- If we add a contact structure, we increase the problem of contacts competition.
- Now the infected individuals may not be able to infect as many individuals as the reproductive number says due to the contact structure.

R_0 estimation in networks

- The attack rate method does not work in this scenario.
- Very poor estimations for all R_0.

R_0	Method		
	Attack rate	Exponential growth	Maximum Likelihood
1.25	1.05	1.12	1.10
1.75	1.23	1.63	1.60
2.50	1.45	1.59	1.30
3.00	1.56	1.27	1.08
4.00	1.77	0.97	1.01

TAKE HOME MESSAGE

- Do not use R_0!!!